

DevOps Lab Programs

Program-3: Code a Dockerized Python Flask

Project Structure:

Program-3/

→ Dockerfile

→ requirements.txt

→ app.py

Steps for Execution:

Step 1: Create a Dockerfile without any extension and add the following content

Nano Dockerfile

```
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docke
GNU nano 8.1 Dockerfile
FROM python:latest

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 5000

CMD ["python", "app.py"]
```

Step 2: Create a app.py file with the following code

```
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/program3
GNU nano 8.1 app.py
from flask import Flask

app=Flask(__name__)

@app.route("/")

def home():
    return "hello"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

Step 3: Create requirements.txt file



```
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:
GNU nano 8.1 requirements.txt
flask
```

Step 4: Execute the Dockr Build command

```

1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/program3$ nano Dockerfile
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/program3$ nano requirements.txt
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/program3$ nano app.py
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/program3$ sudo docker build -t program3 .
[sudo] password for 1rv24mc083-rakshithadas:
sudo: a password is required
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/program3$ sudo docker build -t program3 .
[sudo] password for 1rv24mc083-rakshithadas:
[+] Building 74.2s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 201B
=> [internal] load metadata for docker.io/library/python:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:latest@sha256:f05033a4c0ff84db95fd7e6cb361b940a260703d1cd63c63b3472c8ee48e9cff
=> => resolve docker.io/library/python:latest@sha256:f05033a4c0ff84db95fd7e6cb361b940a260703d1cd63c63b3472c8ee48e9cff
=> => sha256:281b80c799ded5b9a390d2e8c59930db01ee395ab809dd34259897c660751f4e 49.29MB / 49.29MB
=> => sha256:15f14138abe4f09d3ef3953105144728056046ae469ae21e8e42749bacd67cca 25.61MB / 25.61MB
=> => sha256:f05033a4c0ff84db95fd7e6cb361b940a260703d1cd63c63b3472c8ee48e9cff 10.95kB / 10.95kB
=> => sha256:3b700cf7651a764f8c33f574fe4f71be54f77cf626c3b2572e26493ca465e3b1 2.32kB / 2.32kB
=> => sha256:1edf53fea0281306069b0c05371a2ad9c5ac12e7474dc0cd6c4902c7eb33a9c9 6.49kB / 6.49kB
=> => sha256:378c64c4458002be86f2d86c5768ae9ec0cff69afac7d1444e50206dc4566fa9 67.78MB / 67.78MB
=> => sha256:02e37abc533a09f86edcf3602251a61a6208ddaa131e1a1d5757e5f94a54d645 235.98MB / 235.98MB
=> => extracting sha256:281b80c799ded5b9a390d2e8c59930db01ee395ab809dd34259897c660751f4e
=> => sha256:b1741a62cceed4e959c504d7a6a25e533c5783994c386b24d7b31a646163283d 6.08MB / 6.08MB
=> => extracting sha256:15f14138abe4f09d3ef3953105144728056046ae469ae21e8e42749bacd67cca
=> => sha256:add107facb26b82200d59d5550cbbab6af9cf38c971f00a43e3d3cda71ffe76a 29.40MB / 29.40MB
=> => extracting sha256:378c64c4458002be86f2d86c5768ae9ec0cff69afac7d1444e50206dc4566fa9
=> => sha256:1227bf08bd420f8324f592cf370b34de4c11dd57aba693e8f3ec9ccbda8d468d 249B / 249B
=> => extracting sha256:02e37abc533a09f86edcf3602251a61a6208ddaa131e1a1d5757e5f94a54d645
=> => extracting sha256:b1741a62cceed4e959c504d7a6a25e533c5783994c386b24d7b31a646163283d
=> => extracting sha256:add107facb26b82200d59d5550cbbab6af9cf38c971f00a43e3d3cda71ffe76a
=> => extracting sha256:1227bf08bd420f8324f592cf370b34de4c11dd57aba693e8f3ec9ccbda8d468d
=> [internal] load build context
=> => transferring context: 454B
=> [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:beb4ed0236bd03658f1f9f37d9e57d766463ab02d28edcb80927d2b184d4efd
=> => naming to docker.io/library/program3
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/program3$ sudo docker run -p 5000:5000 program3
docker: Error response from daemon: failed to set up container networking: driver failed pre

```

Step 5: Run the docker command and specify the port number

```

1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/program3$ sudo docker images
REPOSITORY          TAG          IMAGE ID          CREATED          SIZE
program3            latest      beb4ed0236bd     About a minute ago  1.13GB
program2            latest      6e2123227237     9 minutes ago    137MB
program1            latest      2dc59dcb6250     14 minutes ago   133MB
rakshithadas/lab6   latest      5f211a0185ab     5 weeks ago      137MB
ghcr.io/rakshithadas/lab6   latest      5f211a0185ab     5 weeks ago      137MB
lab6                latest      5f211a0185ab     5 weeks ago      137MB
prog6               latest      c67f0f30bf8e     5 weeks ago      137MB
rakshithadas/prog6   latest      c67f0f30bf8e     5 weeks ago      137MB
ghcr.io/rakshithadas/prog6   latest      c67f0f30bf8e     5 weeks ago      137MB
program6            latest      b669841c942b     5 weeks ago      137MB
rakshithadas/program6   latest      b669841c942b     5 weeks ago      137MB
ghcr.io/rakshithadas/program6   latest      b669841c942b     5 weeks ago      137MB
p6                  latest      0427065fc4f8     5 weeks ago      137MB
rakshithadas/p6      latest      0427065fc4f8     5 weeks ago      137MB
ghcr.io/rakshithadas/p6   latest      0427065fc4f8     5 weeks ago      137MB
myapp               latest      79481fa1b0ba     2 months ago     132MB
gcr.io/k8s-minikube/kicbase   v0.0.48     c6b5532e987b     3 months ago     1.31GB
hello-world         latest      1b44b5a3e06a     4 months ago     10.1kB
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/program3$ sudo docker run -p 5000
docker: Error response from daemon: failed to set up container networking: driver failed
9d405b31005866a25db9877a3bd4d112c4): Bind for 0.0.0.0:5000 failed: port is already all

Run 'docker run --help' for more information
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/program3$ sudo docker run -p 5000
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.4:5000
Press CTRL+C to quit

```

Step 6: Verify localhost details on browser

